Brendan Birozy

Irvine, CA | (714) 944-7066 | brendanbirozy@gmail.com | linkedin.com/in/brendanbirozy | brendanbirozy.com

EDUCATION

California State Polytechnic University, Pomona **Expected May 2027**

B.S. Electrical Engineering - In Progress

EXPERIENCE

Parker Aerospace, Irvine, CA **May 2026 - Present**

Electrical Engineering Intern, Commercial Flight Controls Division

- Redesign sine-wave inverter driver boards from legacy FPGA-based architecture to both dissimilar FPGA and microcontroller-based control, including schematic/board updates and validation planning.
- Evaluate SiC (Silicon Carbide) power modules for an LLC floating power supply through schematic and board design, circuit analysis, simulation, assembly, and testing.
- Test, analyze, and optimize signal integrity for high-frequency RS-485 and CAN transmission-line cable runs with isolated transceivers.
- Prototype IC implementations on legacy flight-control hardware by modifying existing PCBs with hand-built daughterboard circuitry; validate changes with FCC test consoles.
- Migrate DxDesigner schematics, symbols, and footprints into Altium libraries.

CPP Formula SAE Electric, Cal Poly Pomona **Aug 2025 - Present**

President, Chief Engineer *May 2026 - Present*

- Lead overall vision, strategy, and execution for 50-member Formula SAE Electric team; coordinate people, priorities, resources, timelines, budget/schedule decisions, and compliance.
- Guide overall technical direction across all vehicle architectures while representing team to sponsors and campus engineering community.

Electrical Design Engineer *Aug 2025 - Present*

- Design, manufacture, test, validate, and iterate HV 400V and LV 12V/5V/3.3V boards for Battery Management System (BMS), including shutdown, precharge/discharge, indicator light, sensor, and embedded CAN/SPI/UART circuitry with Teensy4.1.
- Integrate PCBs into 96s4p tractive battery (six 16s4p Molicel P50B modules) and tractive system with Cascadia PM100DX inverter and Emrax 228MV motor as part of 5-member powertrain subteam and 30-member engineering team.
- Develop and build high- and low-voltage wiring harnesses across the vehicle.

System Lead, Business and Marketing *Sep 2025 - May 2026*

- Lead sponsorship strategy, identify and secure industry/alumni partnerships, and draft business proposal and cost analysis submissions with engineering leads to maximize cost-to-performance ROI across subteams.
- Deliver spoken pitch decks to competition judges and help team place top 50 in business event; manage Framer website, sponsor communications, campus engagement, and LinkedIn/Instagram brand assets using Adobe, Canva, and Blender.
- Grow Instagram following from 250 (Oct 2025) to 1,200 (July 2026) while supporting recruitment, retention, and team photo/content production.

Autonomous Vehicle Laboratory, Cal Poly Pomona **Aug 2025 - Mar 2026**

Embedded Systems Team Member

- Design and migrate drive-by-wire steering, throttle, and braking control systems prototypes to permanent PCBs.
- Work with 20-person team to integrate steering, throttle, and braking actuators and motors over CAN using RP2040s and Teensy4.1s; research 4WD hub motor powertrain strategies for off-road autonomous EV applications.

SKILLS

- Hardware: PCB Design, Schematic Capture, Circuit Analysis, Microcontrollers, Microelectronics, Power Electronics, High Voltage/High Current Design, Analog Design, Digital Design, Mixed Signal Design, ADC, FPGA, Digital Logic, Embedded Systems, Soldering, Lab Equipment, Fault/Failure Analysis, Signal Integrity
- Tools/Software: Altium, KiCAD, OrCAD, PSpice, LTspice, MATLAB, Simulink, NI LabVIEW, Verilog, C/C++, CAN, SPI, UART, RS-485, SolidWorks, 3DExperience, Microsoft SharePoint, Microsoft Office, Google Suite, Adobe Suite
- Battery Management Systems, Digital Marketing, Project Management, Technical Communication, Cost Analysis, Team Leadership